

Compatible-Frame Oxygen-Ring Geometry Specification

Stage 11C2: Fixed-Identity Ring Dynamics, Reference-Aligned Frames, and Explicit Unresolved States

mdstats

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Purpose and stage boundary

Stage 11C2 maps the immutable Stage-11C1 ring identities onto the same compatible frames already certified by Stage 11B:

```
reference ring catalog + compatible-frame tiling catalog
  + atomistic frame collection + atomic connectivity
  -> replay fixed T atom/image identities
  -> replay fixed O bridge identities and T-O-T closure
  -> instantaneous O-ring center, plane, and descriptors
  -> reference-aligned two-sided local frames
  -> explicit per-ring and per-frame resolution states
```

Runtime/API target:

mdstats 0.19.88a0

Primary module:

mdstats/analysis/ring_geometry_frames.py

Stage 11C2 is descriptive geometry. It does not assign ions, infer metastable states, label LTA cage semantics, smooth trajectories, repair topology, or fit rates.

Source hierarchy

The Stage-11C1 ReferenceRingGeometryCatalog remains the authoritative source of persistent window, primitive-ring, T-atom, O-atom, and side identities. FrameTilingGeometryCatalog remains the authoritative source of frame compatibility and the periodic projected-framework gauge. Stage 11C2 must not rediscover a different gauge.

The following digests must agree before mapping begins:

- the natural-tiling geometry digest;
- the periodic-cell-complex digest;
- the selected collection/frame binding used by Stage 11B;

- the atomic-connectivity binding used by Stage 11B; and
- the Stage-11C1 reference-frame fingerprint.

Public API

```
map_ring_geometry_to_frames(
    reference_geometry: ReferenceRingGeometryCatalog,
    frame_tiling_geometry: FrameTilingGeometryCatalog,
    collection: AtomisticFrameCollection,
    connectivity: AtomicConnectivityResult,
    *,
    options: FrameRingGeometryOptions | None = None,
    resources: FrameRingGeometryResources | None = None,
) -> FrameRingGeometryCatalog
```

The mapped frame set is exactly the ordered frame set stored by `frame_tiling_geometry`. Frame selection is therefore performed once, upstream, by Stage 11B.

Fixed-identity polygon replay

For a mapped Stage-11B frame, let the canonical projected-framework coordinate of T atom i be

$$\mathbf{f}_i^{\text{can}}(t) = \mathbf{f}_i^{\text{wrap}}(t) + \mathbf{g}_i(t) + \mathbf{G}(t),$$

where $\mathbf{g}_i(t)$ and $\mathbf{G}(t)$ are the stored Stage-11B vertex and global gauges. The lifted T coordinate for reference ring atom/image record $(i, \mathbf{n}_i)$ is

$$\mathbf{f}_i^R(t) = \mathbf{f}_i^{\text{can}}(t) + \mathbf{n}_i.$$

The O atom identity is never reselected. For each fixed T-O-T bridge, the instantaneous minimum-image T-to-O shift is replayed from the lifted source T. The O-to-target-T leg must close onto the already mapped lifted target T within `path_closure_tolerance`. The current connectivity state must contain both fixed T-O and O-T atom pairs. Otherwise the ring receives an explicit missing-bridge or gauge-failure state. Under the current Stage-11B graph-digest contract, a changed framework-linker identity normally makes the whole frame `topology_mismatch` before per-ring replay begins. The `missing_bridge` state is therefore a defensive distinction for compatible future upstream policies or other source bindings that preserve the projected graph identity while a fixed bridge pair is unavailable.

Instantaneous geometry

The O polygon uses the same Pearson closest-fit plane and projected polygon area-centroid definitions as Stage 11C1. Each resolved frame stores:

- lifted T and O fractional and Cartesian polygons;
- O-vertex and O-area centers;
- ordered unit normal;
- two inward side frames;
- covariance eigenvalues;
- vector-area magnitude and projected area;
- perimeter;
- RMS and maximum planarity deviations;
- puckering amplitude;
- ellipticity;
- center aperture radius; and
- both ordered T-O distance arrays.

The O-area centroid remains a geometric center, not an ionic-site center.

Reference-aligned orientation

The ordered normal is sign-aligned to the Stage-11C1 ordered normal:

$$\mathbf{n}(t) \cdot \mathbf{n}_0 \geq 0.$$

The in-plane frame uses corresponding O atoms. First, the reference basis is transported to the current plane by the minimum rotation that maps $\mathbf{n}_0$ to $\mathbf{n}(t)$. Then a two-dimensional proper orthogonal Procrustes fit determines the in-plane rotation that best aligns the reference and instantaneous projected O polygons. This is a descriptive orientation convention, not a dynamical smoothing filter. The orthogonal Procrustes method follows Schönemann [3].

The result stores:

$$\theta_{\text{tilt}}(t) = \cos(\mathbf{n}_0 \cdot \mathbf{n}(t)),$$

and the signed in-plane rotation relative to the transported reference basis.

Translation and deformation descriptors

For each resolved ring, Stage 11C2 reports:

- Cartesian center translation $\mathbf{c}(t) - \mathbf{c}_0$;
- fractional center translation in the current cell gauge;
- normal-reference dot product;
- tilt angle;
- signed in-plane rotation; and
- instantaneous geometric descriptors listed above.

No affine-cell correction is silently applied to the Cartesian translation. Consumers requiring strain-removed motion must derive it explicitly from stored fractional centers and cells.

Resolution states

FrameRingGeometryStatus distinguishes:

- mapped;
- reference_unresolved;
- topology_mismatch inherited from Stage 11B;
- missing_bridge for absent fixed T–O/O–T connectivity;
- gauge_failure for unreplayable integer-image closure;
- degenerate_geometry for an invalid instantaneous O polygon; and
- upstream_frame_unresolved for other unresolved Stage-11B frame states.

Persistent ring identity is retained in every record. Partial frames are allowed: one ring can be unresolved while other rings remain mapped.

At frame level:

- mapped means all reference-resolved rings mapped;
- partially_mapped means at least one but not all reference-resolved rings mapped; and
- unresolved means no reference-resolved ring mapped.

Resource policy

Resource preflight occurs before mapping and bounds:

- selected frame count;
- ring count;
- total T/O vertex instances; and
- minimum-image pair tests.

No partial catalog is returned after a resource rejection.

Immutability, serialization, and metric series

All records are frozen and use immutable tuples. The catalog digest is canonical. `from_dict()` rebuilds the catalog from scientific sources and rejects any noncanonical or tampered payload.

`ring_metric(window_index, metric)` returns a read-only frame-aligned NumPy array and inserts `NaN` for unresolved ring frames. Supported scalar metrics include area, perimeter, aperture, planarity, puckering, ellipticity, tilt, and in-plane rotation.

Validation requirements

Focused validation must include:

- all 58 LTA rings mapped on an unchanged compatible frame;
- equality with Stage-11C1 geometry on the reference frame;
- integer atom-wrapping invariance;
- trajectory-anchor translation continuity;
- isotropic scaling of lengths and areas;
- rigid rotation of centers, normals, and side frames;
- stable reference-aligned orientation under small deformation;
- partial-frame retention under one-ring degenerate geometry;
- topology-mismatch precedence for changed framework-linker identity;
- inherited topology mismatch;
- explicit gauge and degenerate-geometry failures;
- frame and ring resource preflight;
- deterministic metric series;
- canonical replay and tamper rejection; and
- regression of Stages 11A, 11B, and 11C1.

Deferred boundary

Stage 11C2 does not implement framework semantic labels, species-dependent site states, ion assignment, residence statistics, transition networks, or rate models. Those remain Stages 11D and 11E.

References

- [1] Pearson, K. (1901). *On Lines and Planes of Closest Fit to Systems of Points in Space*. Philosophical Magazine, Series 6, 2(11), 559–572. DOI: 10.1080/14786440109462720.
- [2] O’Rourke, J. (1998). *Computational Geometry in C*, 2nd edition. Cambridge University Press.
- [3] Schönemann, P. H. (1966). *A generalized solution of the orthogonal Procrustes problem*. Psychometrika 31, 1–10. DOI: 10.1007/BF02289451.